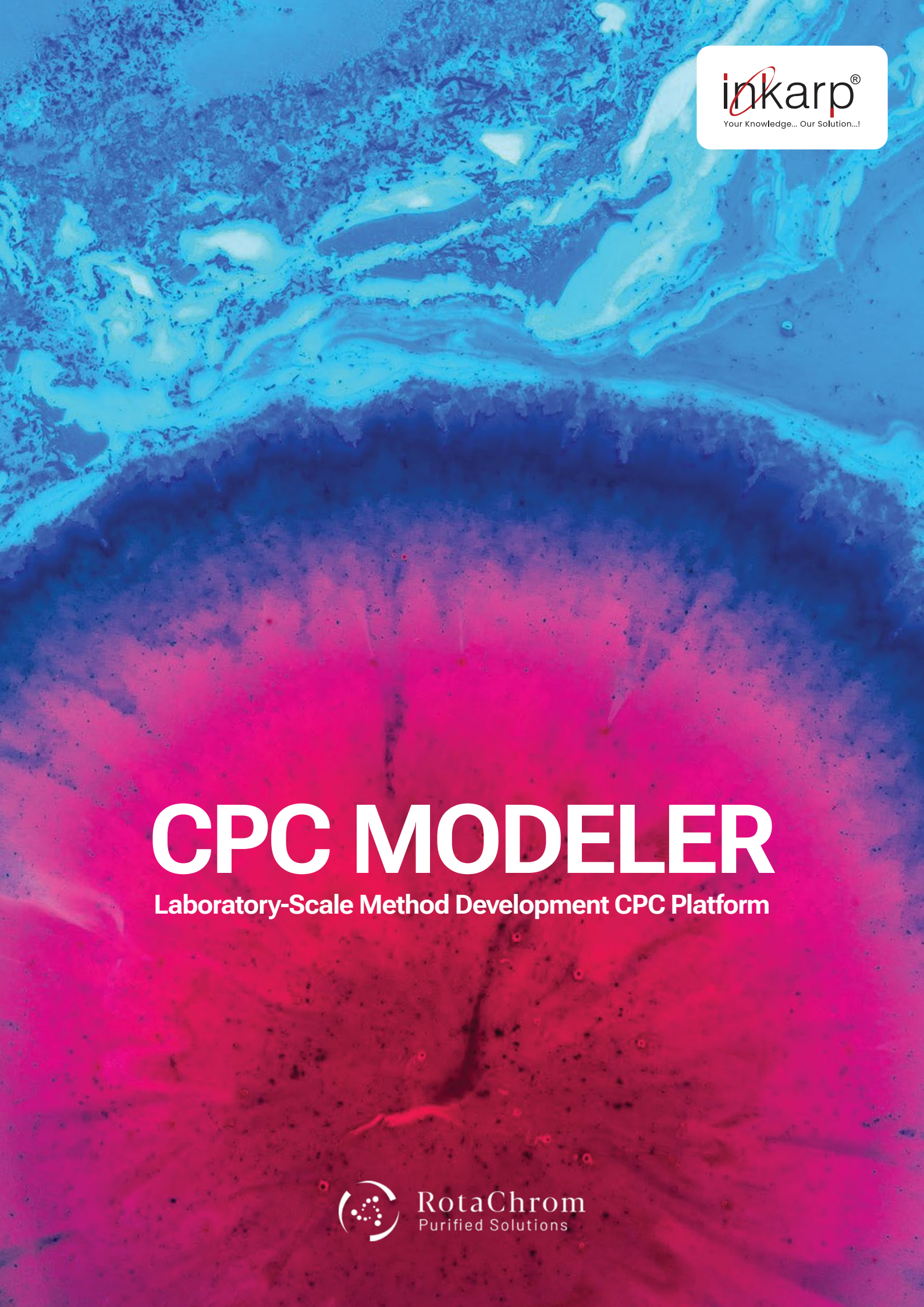

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Your Knowledge... Our Solution...!
CPC MODELER
Laboratory-Scale Method Development CPC Platform
RotaChrom
Purified Solutions

From Laboratory to Production With a Single Technology

A Single Handshake For Long-Term Success

Purifying challenging compounds doesn't just happen by purchasing a purification device and then going into production. Research and development (R&D) and scale-out often provide difficulties and challenges for manufacturers.

Companies often must purchase a variety for devices, which will then be used in just one single step in the downstream process. Current research and development workflows require a method just for the research and development phase, and then need more devices or completely new technologies for the pilot-scale and industrial-scale, with the ones used in the previous steps becoming obsolete later.

However, researchers and producers could use technologies from the very beginning of the R&D phase, which then can be carried over into even industrial-scale production. This means they won't have to invest time and manpower into learning the ins and outs of a single technology just to leave it behind in the next stage, and then keep learning new techniques every step of the scale-out process going into production.

Centrifugal partition chromatography (CPC) offers a solution which can be implemented in your downstream chain right at the beginning, addressing all scale-out and complicated method development issues. Since CPC is especially capable of scale-up efforts, introducing a CPC device during the R&D phase means you can carry it over into pilot-scale research, and them into industrial-scale production. From method development to industrial production, CPC can provide a trustworthy solution for the whole downstream process.

Besides the benefits of an easy scale-out, CPC excels in purification tasks such as separation, isolation and remediation, and can be used for a variety of compounds including APIs, natural extracts, isomers, peptides and chiral molecules.

RotaChrom's CPC Modeler is designed for R&D, method development, solubility and separation testing, and laboratory-scale production. The applicability of CPC technology from research to large-scale production ensures a smooth and easy technology transfer, starting with the CPC Modeler and extending to the largest CPC instruments developed and commercialized by RotaChrom.

Centrifugal Partition Chromatography, Explained

CPC is a liquid-liquid preparative chromatographic technique, in which separation occurs between two immiscible liquid solvent systems: acting as both the stationary and mobile phases. Separation of the injected sample is based on the sample component's varying partition coefficients between the mobile and stationary phases. Partition coefficient (K_d) is the equilibrium constant for the distribution of a compound in a two-phase system.

A pattern of CPC cells (columns) are interconnected in series by ducts attached to a large rotor. This system is then filled with the liquid stationary phase, which is immobilized inside the rotor-cell system by a strong centrifugal force. The other phase of the two-phase system acts as the mobile phase containing the sample to be purified. This liquid is fed under pressure into the rotor and pumped through the stationary phase in the form of tiny droplets.

The column in a CPC device is essentially the rotor and its interconnected cells. This is the feature which gives way to the different methods of operation. Namely, the CPC device can be operated in two different modes: ascending mode and descending mode **(Fig.1.)**. The main factor which decides the mode is the density of the mobile phase.

A liquid mobile phase can be denser or less dense compared to the immobilized liquid stationary phase. When using the 'ascending' mode, the mobile phase is the less-dense phase of the two phases. When this is the case, the direction of the flow becomes the opposite of the centrifugal force. When operating the machine in 'descending' mode, then the mobile phase becomes the denser phase.

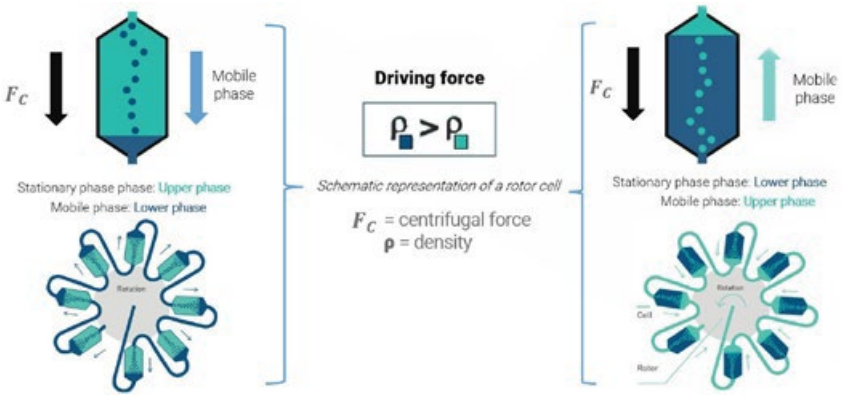


Fig. 1.: CPC cell and rotor operation in ascending and descending modes.

CPC Modeler

Make Your Downstream Journey Easy With CPC

The CPC Modeler is a quick and robust method development platform. It seamlessly integrates an **intuitive method development engine with tailor-made semi-preparative chromatographic equipment**.

Method development takes place in software part of the CPC Modeler, called the CPC Simulator, which is a tool that can predict the optimal solvent system to purify a compound of interest (CoI).

The main hardware in the platform is the Benchtop CPC, which is a small-scale LLC chromatography device, which can test and validate the optimal purification method. The platform is complete with a preparative unit and a fraction collector **(Fig.2.)** to help with chromatographic research and developmet.

The CPC Modeler not only tests, predicts and validates optimal purification methods for various CoIs but also facilitates seamless production scalability.

The Modeler is the smallest member in RotaChrom's ecosystem. Scale-out can be made extremely easy, as the technology offers a solution for every step of the downstream process.

Advantages

- Model and optimize purification performance through advanced method development.
- Speed up your purification method development for increased efficiency.
- Choose from a range of versatile purification systems tailored to a diverse array of CoIs.
- Achieve the highest purity level of your target compounds.
- Tailor your purification process by selecting from lab-scale, pilot-scale, or industrial-scale options.
- Scale-up production seamlessly, from milligrams to kilograms per day.
- Embrace green technology and eliminate silica gel and reduce waste with a flexible solvent handling, recycling, and recovery system.
- Complies with all mandatory regulatory standards and safety requirements for installation, operation, and use in the EU and United States, including CE Declaration of Conformity and UL certifications.



Fig. 2.: Benchtop CPC (left), CPC Simulator (monitor), preparative unit with fraction collector (middle and right).

A Solution for Separation Prediction and Validation

Predict

The CPC Simulator software tool offers easy method-development:

- Conduct a chemical similarity-based query for the Compound of Interest (CoI).
- See the recommendation for an appropriate solvent system and purification method.
- Model and optimize purification performance for enhanced efficiency.
- Provide support for seamless scale-up of purification processes.
- Simulate the outcome of the separation before going to the lab.

Validate

The Benchtop CPC's semi-preparative partition chromatography capabilities:

- Fine-tune operational parameters for optimal performance.
- Deliver purified materials through the chromatographic process.

Benchtop CPC

The Benchtop CPC is RotaChrom's laboratory-scale centrifugal partition chromatography device. It is an integral part of the CPC Modeler platform, and as such, this is the device which carries out the chromatographic separations.

The device is based on RotaChrom's CPC technology, with a customized rotor design outfitted with liquid delivery and detection devices and a fraction collector. As its name suggests, the device can fit on top of a standard desk, making it ideal for laboratory-scale method development tasks.

The Benchtop CPC offers mg to grams scale/cycle loading capacity, and can purify a wide variety of pharmaceutical APIs, natural extracts, stereoisomers and further complex matrices.



Advantages

- Achieve fully integrated method development and solvent screening for RotaChrom's preparative CPC technique. From lab-scale to industrial-scale, CPC will be there to support you every step in your downstream process.
- Meet industry expectations with swift turnaround times facilitated by automated process steps.
- Decrease development time and minimize consumable expenses.
- Streamline method development with a customizable User Interface for enhanced flexibility.
- Support drug discovery for compounds sourced from natural and biotechnological origins, navigating through complex matrices. Additionally, explore uncharted chemical spaces for drug-like target molecules.

Benefits

- Model and optimize purification performance for fast and reliable method development.
- Ensure successful and easy method development and scale-up with one purification platform and technology. No need to buy different technologies for each downstream step.
- Easy, straightforward scalability thanks to RotaChrom's cell design and CPC's nature.
- Large variety on solvent systems and orthogonality.
- Select between versatile purification and solvent systems to work with a wide range of COIs.
- Decrease your expenses due to solvent and stationary phase use, thanks to CPC's liquid-liquid separation techniques, produce less waste with a versatile solvent handling, recycling, and recovery system.

Novel Technology: Cell Design and Geometry

RotaChrom engineers developed a highly efficient, novel CPC cell design used in all our devices. The chromatographic cells are an integral part of CPC rotors, as cell geometry greatly affect the details of the separations.

RotaChrom's cell geometry is designed with so-called Computational Fluid Dynamic (CFD) modelling. CFD ensures that our cells are designed intentionally, and that scale-up efforts can be planned ahead during these cell design stages.

The CFD tool helped our engineers simulate the efficacy of mass transfer between the mobile and stationary phases and predict and expect the plate number outcome of the separation based on the physical and chemical properties of the droplets in the cell, and maximize cell and purification capacity.

RotaChrom devices all use CFD-designed cells, which makes scale-up predictable and easy. Anything you develop with the help of the CPC Modeler can be carried over to pilot-scale and industrial-scale purification with expected results.

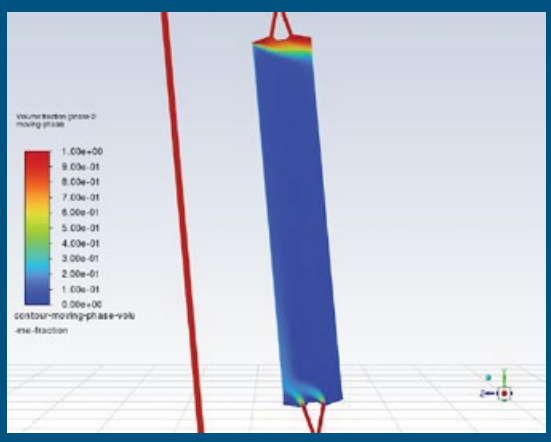


Fig. 3.: The CFD tool can simulate droplet movement inside the cell and help understand the separation process, therefore maximize efficiency.

CPC Simulator

A Tool For Proposing the Optimal CPC Method



Achieving optimal resolution is paramount in developing a purification method. The CPC Simulator tool, as the part of the CPC Modeler, and as an individual tool, lets you find the most effective CPC purification method, visualize chromatographic runs in-silico, suggest a solvent system for your particular compound of interest (CoI), evaluate the partition coefficient, and evaluate the productivity of your CoI.

The core of the CPC Simulator consists of two software tools:

- A database with a chemical similarity-based search engine that contains extensive data from scientific literature, providing purification methods tested with CPC.
- The CPC Visualizer tool, which generates visual representations of method results based on user input.

The CPC Simulator can determine which solvent system will separate your compound of interest from the crude material, and estimate the partition coefficient, which then can be used to simulate the separation process.

The Simulator lets you select between predefined instruments and separation modes. Several important factors affecting peak shape and separation performance are considered in the model calculations to improve the simulator's predictivity. These include high concentration, low solubility, and other factors. By changing a parameter's value, you get instant feedback on how it affects separation. As a result, the tool can give you the elution profile with a more precise estimation than a single partition coefficient value.

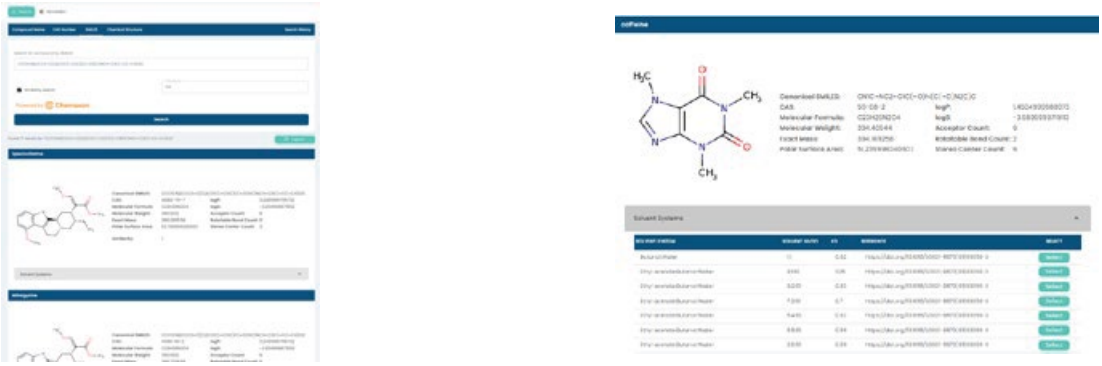


Fig. 4.: The CPC Simulator can search compounds based on chemical similarity (left) and display relevant data referenced by scientific literature (right).

Features and Benefits

Features

- Effective solvent system and purification method development for CPC.
- Only two input data needed for operation: partition coefficient and composition of the crude material.
- Purification performance can be modelled, tested and optimized.
- Thousands of compounds can be tested in silico to find future directions.
- A library of data from scientific literature available, combined with an intelligent search engine, developed by chemists and chemo-informatic experts, highly acknowledged in chromatography.

Benefits

- Quick and cost-effective computational tests instead of slow, cumbersome and expensive experimentation.
- A library of extensive, well-researched and evolving data, instead of time-consuming literature search.
- Precise elution profile with estimation.
- Prediction-based results to support decision making.



Fig. 5.: The CPC Simulator lets you customize several purification details to tailor the simulation.

CPC Visualizer

The CPC Visualizer is part of RotaChrom's predictive tool. It generates a visual representation of a CPC run based on data from the CPC Simulator tool, by utilizing built-in separation scenarios and predefined instrument settings to get a better understanding of the concepts of the examined CPC separation.

After inputting the composition of a crude material and the partition coefficients in a selected solvent system, the application helps you predict if the planned CPC separation will be successful or not without using any laboratory equipment.

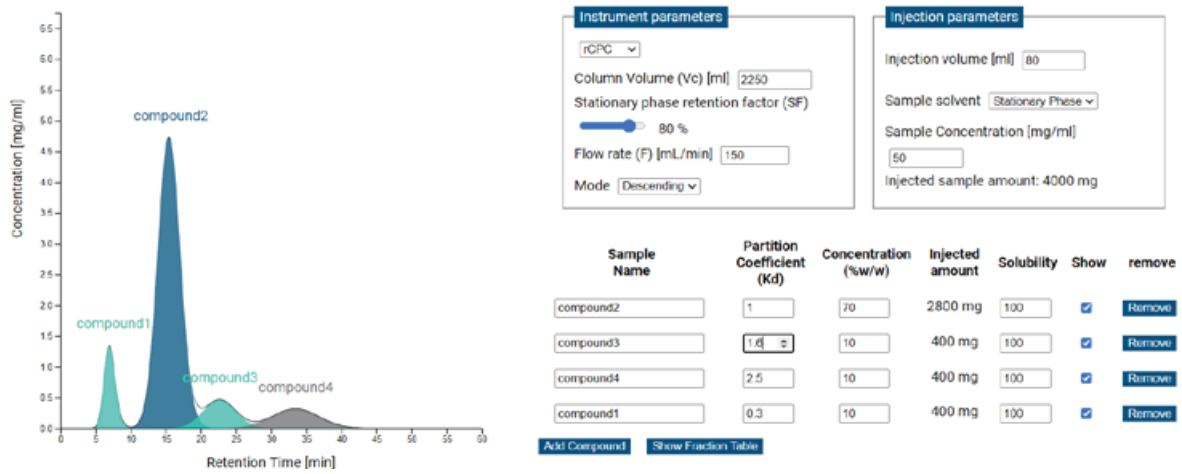


Fig. 6.: Data from the CPC Simulator can be visualized.

A Library Of In-Depth Purification Data

The CPC Simulator gives you access to a database based on 16000 entries of scientific publications, solvent system examples, compound data and in-house research to help you develop the right solvent system and method for your purification goals.

The database lets you to simulate the purification results of 2700 compounds. Each compound comes with several different solvent system combinations allowing you to choose the one most suitable for you purification circumstances and needs.

The database contains an exceptionally large library about natural extracts and compounds. Besides the publications and research, the CPC Simulator's database comes with a detailed API set, so you can simulate and visualize API purifications.

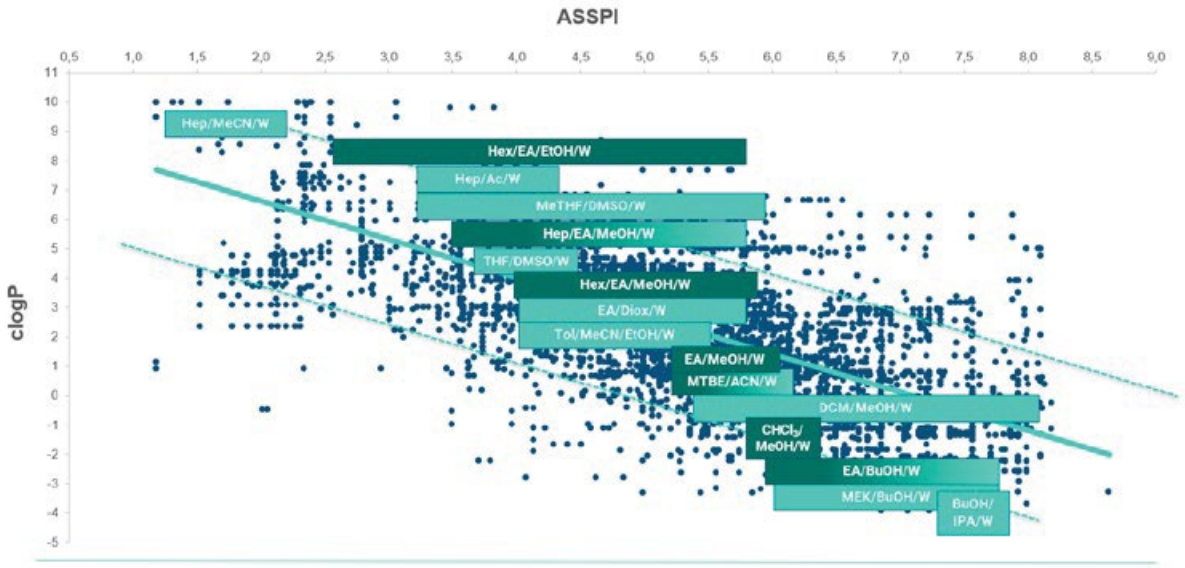


Fig. 7.: The database contains information about common CPC solvent systems and compounds that can be purified with them. Solvent system polarity is marked on the ASSPI scale, while molecule polarity is marked on the logP scale. The points on the diagram show matching points where a solvent system can successfully resolve a compound based on scientific data.

CPC Modeler Platform Technical Data Sheet

RotaChrom Purified Solutions

CPC Modeler Data

Used materials	Ceramic, Teflon, PEEK, glass (FDA/cGMP compliant), SS quality: AISI 316L - EN 10088-1: 1.4404
Interchangeable rotor	Max. diameter 300 mm
Rotor material quality	SS: MSZ EN 10088-1: 1.4404; PEEK
Rotor empty weight	Max. 30 kg
Column capacity	100-250 mL
Flow rate	1-20 mL/min
Rotor Change	Easy/quick
Basic cell geometry	Symmetric, cylindrical cells
Flushability	Easy
Rotation speed	Adjustable, between 700-2400 rpm
Power supply	Separate versions for EU and US standards
Fluid connection points	For 1/16" tubes

Rotor Unit Dimensions

WxDxH	560 x 535 x 410 mm
Empty weight without rotor	63 kg

Machine Specifications

Throughput capability	Grams/cycle
Typical working pressure	80-100 bar
Max pressure	Up to 200 bar
Expected resolving power	N>500
Centrifugal force	>250 g

Standards

Design standards	ASME/BPE and GAMP, CFR 21 Part 11 compliance
Sanitary design	316L SS wetted flow path, FDA conforming non-metallic parts
Electrical and safety approvals	CE/UK CA/NRTL

Product Portfolio

RotaChrom Purified Solutions

RotaChrom's CPC technology offers an easy scale-up and scale-out solution to the partners. Our product portfolio covers solutions for purification, method development, pilot-scale research, continuous production and industrial-scale separation, isolation and remediation.

RotaChrom instruments are available as both standalone instruments, and as part of purification platforms. Our proprietary automated biphasic solvent system handles solvent usage, capable of solvent preparation and recovery and regeneration. All our devices can be constructed according to cGMP compliant standards on-demand.

CPC Platforms and Solutions

CPC Modeler: Purification method development platform

The CPC Modeler is a desktop-sized CPC platform, which combines an intuitive method development engine with tailor-made semi-preparative chromatographic equipment. The CPC Modeler predicts, tests, and validates the most effective purification method of different Compounds of Interest (CoI), while also offering the possibility to scale-up production. The platform offers a mg to g/cycle loading capacity and can purify all the compounds that the rCPC can, including pharmaceutical APIs and natural extracts.



CPC Modeler	
Rotor volume	250 mL
Nominal Flow Rate	1-20 mL/min
Throughput	grams/day
Resolving Power	N>500
GMP Compliance	GMP Compliant Device
Main Feature	In Silico Simulation Tool



rCPC: Pilot-Scale Purification

The rCPC platform is designed to execute lab-scale purification tasks with CPC technology. With its 10g to 100g/cycle loading capacity and potential throughput up to 2 kg/day, it is ideal for all batch sizes, smaller or larger, or campaign types of research work, and is the foundation of our continuous-batch production processes. The device is also available for pharma GMP environment and in URS-compliant equipment design.

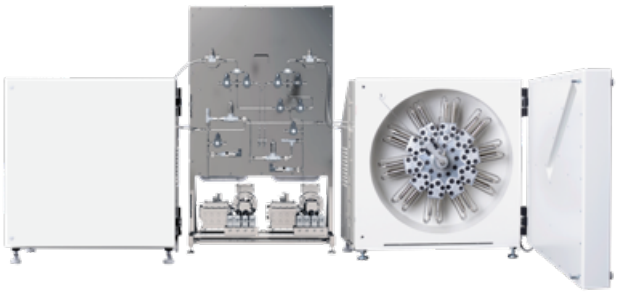
rCPC	
Rotor volume	2.2 L
Nominal Flow Rate	100-300 mL/min
Load Capacity	Up to 150g/injections
Typical Run Time	20-40 min
GMP Compliance	Pharma GMP compliant (Eudralex Volume 4.)

iCPC: Industrial-Scale CPC Purification

The iCPC is the largest serialized CPC instrument in the world, designed for industrial-scale purification projects. This device is by far the most cost-effective industrial-scale chromatography platform available on the market. With its kg/cycle loading capacity, it has the capacity to process up to tens of kg of crude input material per day. The iCPC, also a core element of the pCPC platform, always comes with solvent preparation and recovery and regeneration systems.



iCPC	
Rotor volume	23 L
Nominal Flow Rate	Up to 420 mL/min
Loading Capacity	Up to 1.5 kg/injection
Typical Run Time	20-40 min
GMP Compliance	Pharma GMP compliant (Eudralex Volume 4.) design
Compliance	ATEX Zone 2 /C1D2 compliant design



Continuous CPC: Maximized Productivity With the Smallest Footprint

Continuous CPC delivers purification and polishing solutions for difficult matrices, where the difference between individual constituents is negligible. The platform can operate in multiple-dual mode, which greatly increase production output. The dual rotor design enhances productivity and further reduces the relative cost of consumables using two rotors working together under a single control unit. The platform can be extended with a tailor-made solvent handling system on-demand.

Continuous CPC	
Rotor volume	4.4 L
Nominal Flow Rate	up to 420 mL/min
Sample Injection	Continuous injection of sample (up to 260 ml/min)
Operation Mode	MDM/Trapping MDM/Continuous injection
GMP Compliance	Pharma GMP compliant (Eudralex Volume 4.) design
Compliance	ATEX Zone 2 compliant design

pCPC: Industrial-Scale Isolation, Purification, Separation and Remediation

The pCPC is a combination of the iCPC and a proprietary automated biphasic multi-functional solvent system skid, capable of solvent preparation and recovery and regeneration, providing a flexible and continuous feed to the iCPC instrument. The pCPC's solvent system makes the platform capable of separating, isolating and remediating components from complex matrices to exceptionally high purity.



pCPC	
Rotor volume	23 L
Nominal Flow Rate	2-4 L/min
Throughput	10+ kg/day
Automation & Control	Fully automated industry control system for 24/7 operation
Solvent System	Integrated solvent and sample handling system
GMP Compliance	Pharma GMP compliant (Eudralex Volume 4.) design upon request
Compliance	ATEX Zone 2 /C1D2 compliant design

Chromatography Across the Purification Landscape

Chromatographic methods can be categorized into analytical and preparative, depending on the scale and purpose of the separation process. Analytical methods are designed to provide qualitative and quantitative insights into the sample at hand.

On the other hand, preparative separations aim to separate, isolate or remediate the compound of interest (CoI) from a mixture. These preparative systems can handle large input quantities, processing hundreds or even thousands of grams.

Devices for Preparative Chromatographic Purification

Chromatographic separations can be done on a variety of tools and devices which users need to consider:

High-Performance Liquid Chromatography

High-performance liquid chromatography (HPLC) is a widely used technique for the analysis of purified compounds. It uses a liquid mobile phase and a solid stationary phase to separate and purify compounds in a mixture and a high-pressure pump to carry out the process.

CPC offers several advantages over HPLC:

- Reduced Sample Loss: no irreversible adsorbtion on the Silica stationary phase, no secondary reaction due to Silica. 100% sample recovery throughout the entire process.
- Solvent Versatility: CPC's wider solvent range overcomes HPLC's limitations in separating challenging compounds.
- Scalability: CPC easily scales without major adjustments, suitable for small to large-scale processes, unlike HPLC.
- Cost-Effectiveness: CPC does not require expensive solid stationary phase, neither needs DAC technology to compensate stationary phase deterioration. Parall RotaChrom's unique integrated solvent handling solution allows solvent recycling, greatly reducing expenses.
- Reduced Run Times: CPC achieves shorter run times due to improved mass transfer, compared to HPLC in certain cases.
- Eliminated stationary phase performance degradation: CPC refreshes stationary phase at every single cycle, ensuring stable and comparable purification performance.

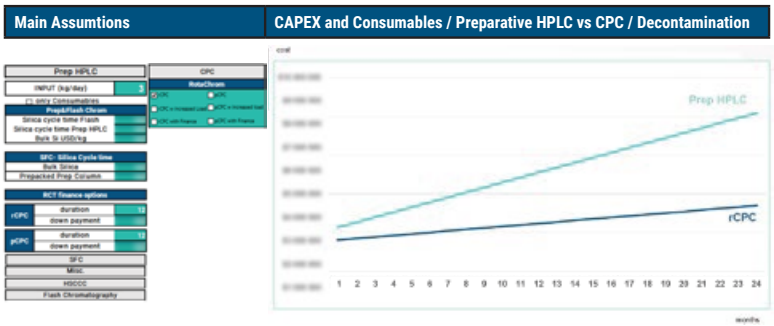


Table 1.: CPC offers lower operational expenses than Prep HPLC over 24-months (expenses are also depend on the type of the purified compound).

Supercritical Fluid Chromatography

Supercritical fluid chromatography (SFC) uses a supercritical fluid as the mobile phase to separate compounds. SCFs are fluids which are above their critical temperature and pressure, and exhibit properties of both liquids and gases. SFC can be particularly useful in cases where the same system of compounds is being purified over an extended period.

CPC's set of advantages can surpass SFC in many fields, including:

- Versatility: CPC accommodates a wide range of solvents for diverse compound types, making it effective for both polar and non-polar separations.
- Scalability: Easily scalable from analytical to preparative scales without major adjustments, suitable for various laboratory sizes and purification processes.
- Cost-effective: CPC requires less expensive equipment than SFC, enhancing accessibility for budget-constrained laboratories, aided by the absence of high-pressure pumps and stationary phase reuse.
- Sample Recovery: CPC's two-phase liquid system minimizes sample loss, allowing for high sample recovery rates and potentially higher yields.
- Minimal Sample Degradation: Operating at lower pressures and temperatures, CPC reduces the risk of sample degradation, particularly advantageous for sensitive compounds.

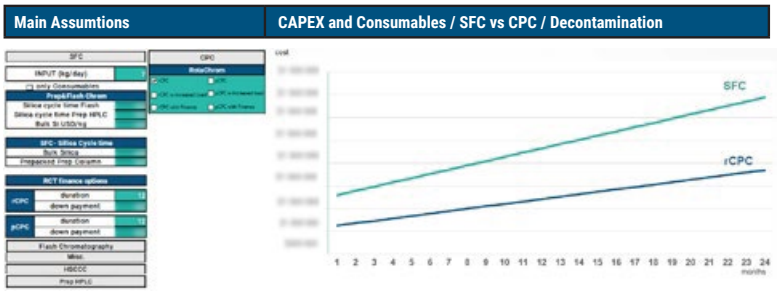


Table 2: CPC offers lower operational expenses than SFC over 24-months (expenses are also depend on the type of the purified compound).

Flash Chromatography

Flash chromatography uses air pressure, or a pump in place of gas to move solvent through a column filled with a solid. Flash chromatography is typically used for medium-scale preparative purification, providing moderate throughput and allowing for the separation of larger quantities of compounds quickly.

CPC surpasses flash with many of its characteristic advantages:

- Higher Sample Loading: CPC efficiently handles larger volumes for purifying greater quantities of compounds.
- Reduced Solvent Consumption: CPC uses less solvent, reducing waste and operating costs, particularly in large-scale purifications.
- Scalability: CPC easily scales from analytical to preparative without significant adjustments, adapting to both small and large-scale processes; Flash Chromatography may need modifications.
- Sample Recovery: CPC's continuous operation yields higher sample recovery rates, leading to increased yields compared to Flash Chromatography.
- Reduced Packing Requirements: Unlike Flash Chromatography, CPC eliminates the need for solid support material and flash cartridges, reducing expenses, setup time and potential packing issues.

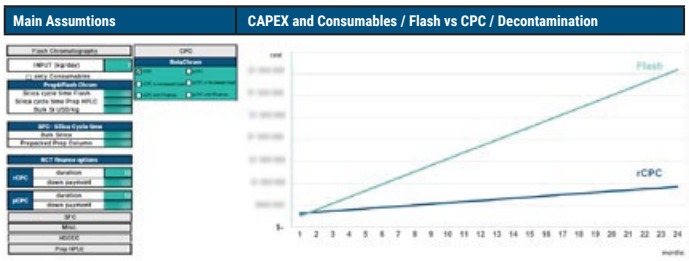


Table 3: CPC offers lower operational expenses than Flash over 24-months (expenses are also depend on the type of the purified compound).

Advantages of CPC

- Scalability: CPC offers a straightforward separation mechanism, making scale-out efforts easy.
- No Sample Loss: CPC's design offers high loadability and ensures optimal recovery, eliminating sample loss attributable to irreversible adsorption.
- No Silica Gel, Reduced Expenses: The liquid-liquid nature eliminates the need for a solid stationary phase, greatly reducing the cost of consumables and enhancing cost-efficiency for sample purification.
- Versatility: CPC can separate a wide array of compounds, encompassing proteins, nucleic acids, and small molecules. This makes CPC a preferred choice across many domains, including biotechnology, pharmaceuticals, and other realms where high-purity samples are essential.

Comparison of Chromatographic Solutions

Aspect/Parameter	preparative LC	CPC
Stationary Phase Retention (S_F)	10-15% Estimated, not adjustable.	50-90% Easily measurable, adjustable
Phase Ratio (V_S/V_M)	0.11-0.15 Volume ratio of stationary/mobile phase is only estimated, cannot be measured	2-6 Volume ratio of stationary/mobile phase is well known, can be measured
Plate Number (N)	>10.000	<500
Peak Width (W)	Narrow	Broad
Peak Capacity (n_{peak})	High (>10) But limited polarity range Only elution mode	Low (<5) But wide polarity range variable operational modes (e.g. elution-extrusion, multiple dual mode, pH-zone refining, etc.)
Selectivity (α)	Typically <1.2 Limited chemistry of stationary/mobile phases	Typically >1.5 Infinite combinations in stat/mob phase Option for multiple dual mode if α<1.2
Loadability (injected volume)	<1% of column volume	Typically min 10% of rotor volume
Robustness	Risk of clogging, irreversible adsorption	Injection of heterogenous sample is sometimes possible

Table 4: CPC offers several advantages compared to preparative LC.

	HPLC	SFC	Flash	CPC
System Phases	Liquid-solid	Supercritical fluid-solid	Liquid-solid	Liquid-liquid
Sample Recovery	Low, sample loss possible	Low, sample loss possible	Low, sample loss possible	High, with low sample loss
Scalability	Difficult	Difficult	Difficult	Easy, linear
Operating/Maintenance Expenses	High	High	High	Low
Solvent Purity Required	HPLC-grade	HPLC-grade	HPLC-grade	Min. technical-grade
Single-Peak Resolution	Yes	No	Yes	Yes

Table 5: CPC can outperform its competitor technologies in sample recovery, scalability, costs of operation and solvent purity.

Centrifugal Partition Chromatographic Purification of Semaglutide

A GLP-1 Receptor Agonist

RotaChrom Purified Solutions

APP-SEMAGLUTIDE-251

Semaglutide **(Fig.1.)** is currently one of the most promising antidiabetic drugs. Approved as a second-line treatment option for type 2 diabetes mellitus, it is highly effective in improving glycemic control and promoting weight loss.[1] It is also under investigation for anti-obesity applications.[2]

Semaglutide is a glucagon-like peptide-1 (GLP-1) receptor agonist available as monotherapy in both subcutaneous and oral dosage forms, with the oral formulation being the first approved oral GLP-1 receptor agonist. It has been optimized for an extended half-life in humans, approximately 7 days, facilitating once-weekly dosing. This prolonged half-life is achieved through structural modifications that enhance albumin binding and confer resistance to degradation by dipeptidyl peptidase-4 (DPP-4).[1,3]

However, semaglutide production remains costly, and achieving high yield, low cost, and high purity continues to pose a significant challenge.

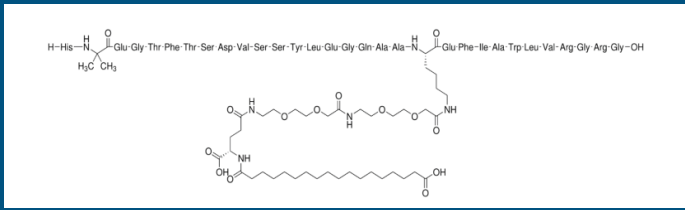


Fig. 1. *Chemical structure of the compound of interest (CoI)*

Key Purification Challenges

- A multi-step purification process, typically involving multiple reversed-phase high-performance liquid chromatography (RP-HPLC) steps using different stationary phases.
- The presence of racemized impurities and sequence variants (structurally similar to the target peptide) pose significant challenges for chromatographic separation and purification.
- A complex impurity profile, including high molecular weight proteins (HMWPs) and a variety of hydrophilic and hydrophobic impurities.
- The need for rigorous impurity control to ensure product safety and efficacy.
- Difficulties during pH adjustment, particularly precipitation near the isoelectric point, along with various stability concerns[4].

Considering the chromatographic complexities and the high financial burden associated with these purification steps, a more cost-effective and promising alternative could be centrifugal partition chromatography (CPC).

Centrifugal Partition Chromatography

CPC is a liquid-liquid chromatographic technique, in which separation occurs between two immiscible liquid phases.

Both the mobile and stationary phases are liquids, and the liquid stationary phase is immobilized by a strong centrifugal force in the cells which are connected by a series of ducts attached to a rotor.

The solute's retention depends simply on the liquid-liquid partition coefficient (K_d). Practically, it is the ratio of the solute concentration in the stationary phase over the solute concentration in the mobile phase.

The Advantages of CPC in the Downstream Process

- No solid stationary phase required
- Gentle separation technique
- Wide compound compatibility
- **Multiple operating modes available (ascending, descending, dual-mode)**
- Flexible method development
- Stable on both acidic and basic pH values

Multiple Dual-Mode CPC (MDM-CPC)

Multiple dual-mode (MDM) separation in CPC consists of a sequence of dual-mode runs, i.e., repeated inversions of the stationary and mobile phases, enabled by the liquid–liquid nature of the system. Each cycle includes two steps: descending (DSC) and ascending (ASC) elution modes, which correspond to reversed and normal-phase operation, respectively.

By switching flow direction and phase roles, MDM enhances resolution and improves the purity of the target compound. It is especially effective for isolating closely eluting or intermediate components from complex mixtures, an advantage unique to liquid–liquid chromatography.

Experiments and Results

To identify the most suitable solvent system, several biphasic solvent systems were evaluated using the shake-flask method to determine ideal partition coefficients for the target compounds. Ultimately, a quaternary organic–aqueous mixture was used, consisting of organic solvents and water. This combination formed a biphasic system with tunable polarity, which made it suitable for peptide separation. The system provided sufficient stability for a scalable CPC run, and the solubility of semaglutide in this mixture was ideal **(Fig.2.)**.

The starting synthetic crude material contained 77—80% semaglutide by weight along with a significant amount of impurities. A total of 900 mg of crude was injected into the Benchtop CPC system operating in multiple dual-mode (method parameters are shown in **Table 1.**). With this specific purification method, the system switches multiple times between descending and ascending modes.

As a result, the process consists of three consecutive segments: DSC, followed by ASC, and then DSC again, representing 1.5 cycles of the MDM elution steps. This continuous switching between two operational modes enhances the selectivity between semaglutide and its impurities, as they migrate at different rates under each elution condition. This results in their separated elution at opposite ends of the rotor.

Semaglutide eluted between 54 and 69 minutes **(Fig. 3)**. Fractions with >95% purity, determined by HPLC-DAD at 254 nm **(Table 2.)**, were pooled and lyophilized. This resulted in a final purity of 98% and a calculated yield of approximately 87%. **Fig. 4.** shows the HPLC chromatogram of the crude sample before CPC purification and the chromatogram of the purified sample below.

Biphasic solvent system screening

The aim is to find a stable solvent system with the right phase ratio and settling time, and in which the sample is completely dissolved.

Determination of K_d ($0,5<K_d<2$)

Shake-flask method was used to determine K_d. The relative concentration of the CoI in each phase was determined by HPLC.

K_d < 0.5: narrow peaks, possible overlap with early eluting compounds

K_d > 2.0: broad peak shape, peak tailing, risk of overlapping with later-eluting compounds.

The most suitable solvent system

The most stable and scalable solvent system, providing the highest solubility for semaglutide, was identified. The injected amount of sample could be further increased to 900 mg.

Multiple dual-mode (DSC-ASC-DSC)

The selectivity between semaglutide and its impurities, such as GLP-1 9–37, dipeptides, the di-deprotected form, Stg-5, the di-protected form, and triphenylmethanol (TrOH), was increased by modifying the elution mode to a multiple dual-mode setup.

Semaglutide purity level exceeds 98%

Fig 2. *CPC method development for semaglutide purification*

Instrument:	Benchtop CPC purification system equipped with a 250 mL rotor
Solvent system:	Organic–aqueous biphasic system
Mode:	DSC-ASC-DSC
RPM:	2500
Flowrate:	10 mL/min
Sample loading:	900 mg injected semaglutide crude dissolved in 10 mL upper phase
Detection:	DAD (254 nm)
Runtime:	80 minutes

Table 1. *The parameters of the CPC method*

Column:	HPLC C_{18} column (150 x 4.6 mm, 2.6 µm)
Mode:	gradient
Flowrate:	0.7 mL/min
Detection wavelength:	254±4 nm
Mobile phase:	A: water +0.1% TFA; B: acetonitrile
Timetable:	0.00 — 3.00 min – 35% B isocratic 3.00 — 20.00 min – 35-80% B gradient 20.00 — 22.00 – 80% B isocratic 22.01 — 25.00 – 35% B isocratic
Injection volume:	10 µL
Column temperature:	40 °C
Diluent and Blank:	MeOH

Table 2. *The analytical parameters of the HPLC method*

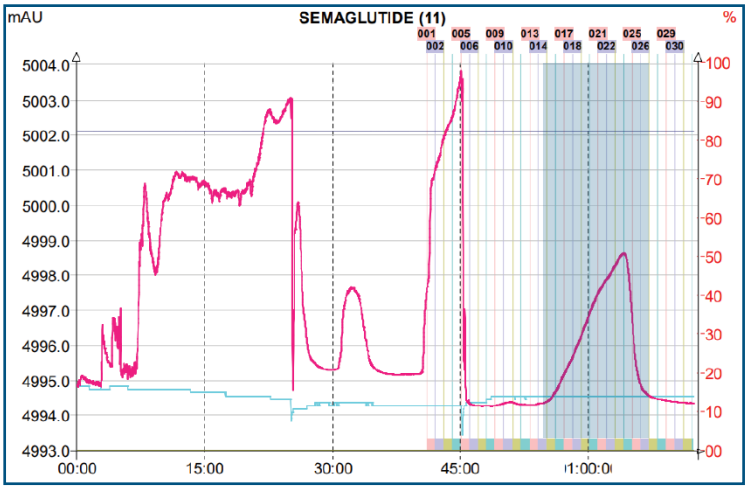


Fig 3. *CPC chromatogram of semaglutide purification (detection: λ=254 nm). Semaglutide peak marked by rectangle.*

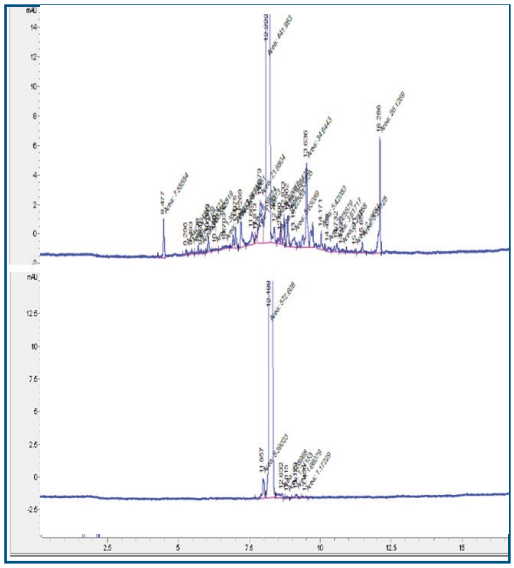


Fig 4. *The chromatogram of the crude sample (top), and the chromatogram of the purified semaglutide fraction (bottom).*

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APPLICABILITY OF CENTRIFUGAL PARTITION CHROMATOGRAPHY IN DOWNSTREAM BIOPROCESSING

RotaChrom Purified Solutions

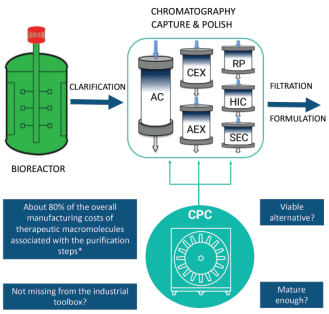
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INTRODUCTION

Chromatography is dominant and inevitable among separation technologies utilized in the complex and multistep purification of therapeutic macromolecules: a myriad of methodologies, such as reversed-phase column chromatography (RP-LC), affinity chromatography (AC), ion exchange (IEX), or size exclusion (SEC) techniques have been widely implemented in the industry. However, it has also become obvious that the limitations of these approaches in terms of efficiency, capacity, compatibility, or sustainability manifest in enormous costs. To provide alternative solutions, significant research and progress have been made in the field of liquid-liquid chromatography (LLC) in the last decade: viability of this approach has been demonstrated on the purification (in capturing and polishing as well) of synthetic oligopeptides, oligonucleotides, enzymes, transport proteins, monoclonal antibodies (mAb), and virus-like particles (VLP).

Centrifugal partition chromatography (CPC) is one realization of LLC, where both the stationary and the mobile phases are liquids and resolution is simply governed by the partitioning of solutes between these two immiscible liquid phases. In practice, one of the two phases is immobilized by a strong centrifugal force inside a rotor, while the other one containing the sample to be purified is pumped through the rotor. Thanks to the liquid nature of the chromatographic system, CPC possesses numerous unique features: (i) flexible and versatile since the chemistry and polarity range of the BLS can be freely varied and adjusted; (ii) total sample recovery can be realized since no irreversible adsorption can occur; (iii) robust and highly loadable due to high stationary phase ratio; (iv) green by appropriate solvent recycling; (v) and can be easily scaled up.

Recent advancements on six relevant fields of biomolecule CPC are summarized in this poster.



*Merck, MK-WP4876EN Ver. 1.0, Vaccine Manufacturing: Collaboration Helps to Overcome Vaccine Process Challenges

SUMMARY

CONS	PROS
ATPS: recycling and recovery (back-extraction)	ATPS: well-established gentle medium with unique selectivity
Slow mass transfer Limited throughput	Specific cell geometries for high stationary phase retention
Unspecific binding Compromised bioactivity	Displacement mode & high injection volume: high productivity
	Robust and easy scale-up

#	Targeted biomolecule	Methodology	Scale	Readiness
1	Peptide	pH-zone, IEX (displacement)	Pilot	Ready for scale-up
2	Proteins	ATPS, IL	Pilot	Scaled-up
3	mAb	ATPS/E, IL	Lab	PoC* ready
4	VLP	ATPS+precipitation	Lab	PoC ready
5	Oligonucleotide	IEX (displacement)	Lab	Applied in production
6	Recycling of ATPS	Ultrafiltration	Lab	Research phase

*PoC: Proof of Concept

Take-home message:
CPC is ready to challenge biomolecule polishing IEX and RP methods even at large scale.

1. PEPTIDES

Traditional processes for peptide purification generally involve a chromatographic step, such as reverse phase high performance liquid chromatography, or ion exchange solid phase chromatography. The liquid-liquid chromatographic "counterpart" of these techniques were developed by the Renault group [1,2]: by dissolving a displacer in the mobile phase and a retainer or an ion-exchanger in the stationary phase, a displacement mode can be performed in sequential fashion. These highly productive methodologies are referred to as pH-zone refining and/or ion-exchange CPC (IX-CPC). IX-CPC was successfully applied for the challenging **polishing** of dirucotide, a synthetic 17-aminoacid peptide in the biphasic methyl tert-butyl ether/acetonitrile/n-butanol/water (2:1:2:5, v/v) solvent system in descending mode using di(2-ethylhexyl)phosphoric acid (DEHPA) as a cheap lipophilic cation-exchanger and Ca2+ ions in aqueous solution as displacer. Critical impurities were efficiently eliminated and dirucotide was recovered in high yield and purity (69% and 98%, respectively) and with a productivity of 2.29 g per liter of stationary phase per hour **(Fig. 1.1)** [3].

In a feasibility study for a generic pharma company, a **capturing** type of peptide purification method was developed in our laboratory. A 30-mer antidiabetic peptide was successfully captured from a fermentation broth beside inorganic ions, organic acids, and polypropylene glycol in the n-BuOH/AcOH/H2O (4:1:5, v/v) solvent system in descending mode at lab-scale. The target peptide was fully recovered (!) in fraction no. 9 and its concentration was increased to about 22-fold in one single run compared to the very dilute level in the crude (0.42 w/w%) **(Fig.1.2.)**.

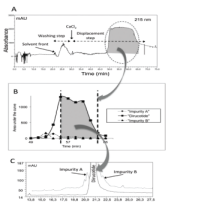


Fig. 1.1. (A) CPC chromatogram of the purification of 197 mg of dirucotide; (B) corresponding HPLC fractogram; (C) HPLC analysis of dirucotide before and after CPC.

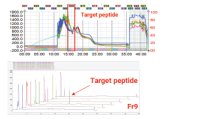


Fig. 1.2. CPC chromatogram and corresponding HPLC traces of fractions.

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2. PROTEINS

Aqueous two-phase systems (ATPS) that are formed by mixing a polymer (usually polyethylene glycol, PEG) and a salt (e.g., phosphate, sulphate, or citrate) or two polymers (PEG/dextran) and water show a proven track record for the extraction and purification of proteins in CPC. Due to their high water-content (>85%), these systems provide an ideal environment for cells, organelles, peptides, and other macromolecules. Partitioning of the protein will be mainly dependent on the size, hydrophobicity, and surface net charge of the protein and the composition, pH, ionic strength, and the presence of additional salt in the ATPS [1,2].

Schweinheer et al. isolated laccases from *Pleurotus sapidus* culture supernatant by CPC at 1000 rpm rotational speed, 5 mL/min flow rate in an ATPS consisting of PEG-3000 (13% w/w)/Phosphate (6% w/w)/NaCl (2.2 or 2.5% w/w). The results of the study showed that the shear and interfacial forces inside the instrument did not lower the activity of the laccases significantly **(Fig.2.1)** [3].

In addition, specifically designed cell geometry for higher stationary phase retention with ATPS [4], systematic ATPS selection [5,6], and ATPS tailoring solutions (e.g., with ionic liquids, see **Fig.2.2**) [7] have been recently published and improved the feasibility and acceptance of this approach.

The human serum albumin isolation study of our lab is presented on the poster of Márió Nacsa (PT-128).

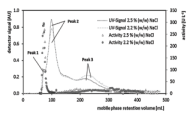


Fig. 2.1. UV-signal and enzyme activity profile of a lab-scale CPC run to isolate active laccases. Descending mode, stationary phase retention ratio was 72%.

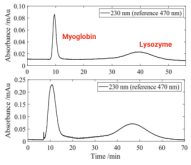


Fig. 2.2. CPC chromatogram of myoglobin and lysozyme separation with a concentration of 2.3 mg/mL for each protein, flow rate of 10 mL/min at 2000 rpm for PEG1000/K$_2$HPO$_4$/KH$_2$PO$_4$/H$_2$O 15.00:9.68:5.32:70.00 (w/w/w/w) with (top) 2.5 wt% [EMIM][Cl], V_{column}=20 mL, initial Sf of 0.67, $P_{myoglobin}$ was 0.07 and $P_{lysozyme}$ 1.84; (bottom) 1.5 wt% [EMIM][Cl], $V_{injection}$=50 mL, initial Sf of 0.72, $P_{myoglobin}$ was 0.2 and $P_{lysozyme}$ 2.11.

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3. MONOCLONAL ANTIBODIES (mAbs)

For the purification of three monoclonal antibodies (mAb) from host cell protein (HCP) precipitation and redissolving, batch partitioning in PEG400/citrate-systems, CPC separation (single and dual mode) in the same type of ATPS, and the combinations of these process steps were investigated by Oelmeier et al. It was found that although mAb recovery was >96% in all cases, neither batch partitioning nor CPC could reduce HCP content satisfactorily. The best results were observed in the case of precipitation and re-solubilization (96.4%). However, when the techniques were combined, the different process steps seemingly removed different population of HCP, resulting in a >98% reduction, while mAb recovery only slightly decreased (~93%). The authors also pointed out that the implementation of the combined process containing a CPC step might yet be problematic in GMP surroundings **(Fig. 3.1)** [1].

Ignatova at Brunel University and coworkers of GSK have recently demonstrated the mAb capturing and fractionation capability of counter-current chromatography/extraction (CCC/CCE). ATPS variables such as PEG molecular weight (400/1000/3350 Da), PEG and salt concentration (14/14 and 20/20% w/w), presence of an additional salt (none, 5% NaCl, 5% NaClO4, and salt type (ammonium sulphate, di-potassium phosphate and sodium citrate) were tested on partitioning of the mAb and impurities. Protein A, SEC, and SDS-PAGE were used to monitor the CCC separation. 200 mg/h throughput and 70% yield was realized in a 1000 mL coil volume in dual mode (bind&elute) **(Fig. 3.2)** [2].

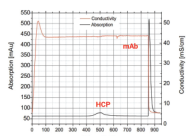


Fig. 3.1. Elution profile of a dual mode CPC run performed with PEG400/citrate-system (21.02% and 18.99% w/w). Operating conditions: 5 mL/min, 2500 rpm. Sample volume: 20 mL. Sample: solved in the upper phase with total concentrations of 5 g/L mAb2 and 0.41 g/L HCP. Elution of HCP after 500 mL was clearly resolved from the elution of mAb after 850 mL.

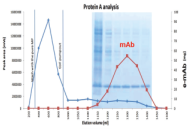


Fig. 3.2. CCC elution profile and protein A map of a mAb sample in dual mode (bind&eluted with a PEG1000/Na-citrate/NaCl system (14%, 14%, 5% w/w). Operating conditions: 10 mL/min, 1400 rpm, 1000 mL coil volume. Run time: 80 min. Sample volume: 225 mL.

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4. VIRUS-LIKE PARTICLES (VLPs)

Virus-like particles (VLPs) represent a new class of bionanoparticles incorporating unique features for medical applications. VLPs are composed of one or several viral structural proteins capable of spontaneously self-assembling into capsids. Thus, they can either mimic the pathogens they derived from or present antigen epitopes of foreign pathogens or tumor cells and can be utilized in recombinant vaccine development. However, especially their complex downstream processing (ultracentrifugation, membrane filtration, IEX, and SEC) faces major challenges, such as low yields and the lack of platform processes: clinical-grade VLP production is still challenging [1].

Ladd Effio and coworkers studied the effect of different ATPS characteristics on the partitioning of human B19 parvo-VLPs derived from *Spodoptera frugiperda* Sf9 insect cells and the removal of host cell contaminants such as cell debris, host cell proteins (HCPs), and DNA. The solubility-guided ATPS optimization revealed important correlations between VLP partitioning and PEG MW and system pH; and shed light on the effect of additional salt on stationary phase retention and VLP/HCP resolution (see **Fig. 4** for optimized solvent system and operating conditions). Nevertheless, the authors concluded to develop an alternative CPC-based multi-stage ATPE process and combine it with a precipitation step to maximize DNA removal (99.77%), VLP purity (90.6%), and yield (63.9%) [2].

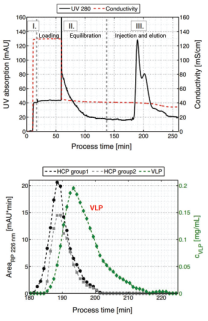


Fig. 4. CPC purification run performed with *Spodoptera frugiperda* Sf9 insect cell lysate containing human B19 parvo-VP2-VLPs. Operating conditions were: 3 mL/min, 2500 rpm. Sample volume: 30 mL. Sample composition: top phase of a PEG400 system (14.08% (w/w) PEG 400, 16.28% (w/w) phosphate pH 8.5, 7.5% (w/w) NaCl) with a total concentration of 0.5 mg/mL VP2-VLP.

REFERENCES
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5. OLIGONUCLEOTIDES

Oligonucleotides comprise relatively short sequences of nucleic acid polymers, which may be DNA or RNA, typically with fifty or fewer bases, and manufactured by solid phase synthesis. With over 300 candidates in clinical trials, oligonucleotides are one of today's most promising therapeutic modalities.

Unfortunately, during the course of synthesis, contamination of the target sequence with shortmer (e.g., N-1 and N-2 base sequences) and longmer (N+1 bases) impurities cannot be avoided. Therefore, extensive RP and/or IEX mode MPLC/HPLC purification steps are necessary, which often provide inferior yield and are problematic in scale-up.

To remedy this, Douillet and coinventors at GSK developed and patented the liquid-liquid chromatography approach of oligo purification. This methodology is based on the well-established ion-exchange displacement mode of CPC (see **Fig. 5.1**) and characterized by high productivity. The technique was validated in a total of 31 different experimental setups on two different lab-scale CPC rotors with protected and deprotected DNA and RNA samples, respectively. In the example, shown in **Fig. 5.2**, purity >94% and yield of 70% were achieved in a 120 min run for a 400 mg crude sample of 89% purity. In another embodiment, 100 mg protected DNA sample with 63% target content was purified to 90% with 68% yield.

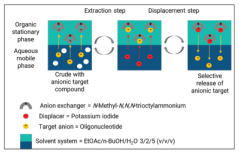


Fig. 5.1. General scheme of ion-exchange (IEX) chromatography exemplified by a solvent system and retainer&exchanger developed and patented for oligonucleotide purification.

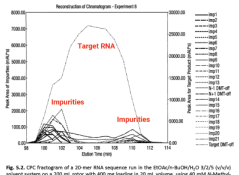


Fig. 5.2. CPC Fractogram of a 20-mer RNA sequence run in the EtOAc/n-BuOH/H$_2$O 3/1/5 (v/v/v) solvent system on a 200 mL rotor with 400 mg loading in 20 mL volume, using 40 mM N-Methyl-N,N,N-trioctylammonium as retainer in the upper stationary phase, 13.3 mM KI as displacer, and 10 mM NaOH in the lower mobile phase. Flow rate 10 mL/min, rotation speed 1200 rpm.

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6. AQUEOUS TWO-PHASE SYSTEM RECYCLING

Santos et al. modelled the continuous separation of PEGylated and non-PEGylated Cytochrome-c (Cyt-c) as model proteins using CPC. By the means of ultrafiltration, the PEGylated product and the unreacted protein could be isolated with high recoveries (88–100%) and purities (~100%) (see **Fig. 6.1**), while the PEG-rich ATPS used in the CPC separation could be recycled and reused in a consecutive cycle of purification without impacting the separation performance (see **Fig. 6.2**). Comparing the process with and without recycling, it was found that nearly a 100% decrease in the E-factor and a 67% reduction of carbon footprint can be achieved using ultrafiltration.



Fig. 6.1. CPC Chromatogram representing the separation of unreacted Cyt-c and Cyt-c-PEG, obtained after two-steps of PEGylation + fractionation in the ATPS comprising 15 wt% of PEG2000 + 20 wt% of potassium phosphate buffer at pH 7.0. Operational parameters: 2.5 mL/min flow rate, 2000 rpm on a 50 mL rotor with 41% stationary phase retention ratio.

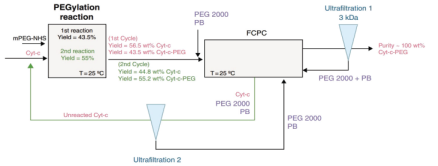


Fig. 6.2. Schematic representation of the solvent system recirculation process by ultrafiltration in the PEGylation model reaction of Cytochrom-c (Cyt-c). FCPC stands for fast centrifugal partition chromatography; PB stands for phosphate buffer.

REFERENCES
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THE CHIRAL SEPARATION OF VORICONAZOLE UTILIZING CONTINUOUS CENTRIFUGAL PARTITION CHROMATOGRAPHY

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INTRODUCTION

Voriconazole is a widely applied antifungal medication. Its synthesis typically results in a racemic mixture of (2*R*,3*S*)- and (2*S*,3*R*)-enantiomers, where the (2*R*,3*S*)-form is the eutomer, while the (2*S*,3*R*)-isomer is the less potent distomer. [1]

Centrifugal partition chromatography (CPC) is a liquid-liquid chromatographic technique where both the stationary and mobile phases are liquids, and the resolution is governed by the partitioning of solutes between these phases. CPC technique for purifying racemic voriconazole via a semi-batch mode (stacked injections) has increased productivity by threefold compared to conventional resolution, while costs are comparable [2]. The dual-mode CPC (MDM-CPC) technique facilitates the simultaneous execution of stationary and mobile phase functions within a single run, thereby offering a streamlined approach for separation.

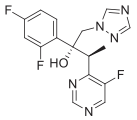
Continuous CPC implements the unique MDM-CPC approach in which two CPC devices are interconnected. Multiple Dual-Mode (MDM) separation involves the alternation between the stationary phase and the mobile phase multiple times, ensuring uninterrupted operation and continuous production.



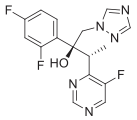
Figure 1: *Dual rotor pilot-scale Continuous CPC device*

MATERIALS & METHODS

Racemic mixture



(2*R*,3*S*)-voriconazole
eutomer - API



(2*S*,3*R*)-voriconazole
distomer

Figure 2: *Voriconazole enantiomers*

CPC method

- Separation occurs between two **immiscible** liquid phases
- Stationary phase is immobilized inside the rotor by a strong **centrifugal force**
- The mobile phase containing the sample to be purified is fed under pressure into the rotor and pumped through the stationary phase in the form of **tiny droplets**
- The chromatographic column in CPC is the **rotor**: cells interconnected in series by ducts attached to a large rotor
- Simple mechanism: difference in **partition (K_D)**
- Continuous operation – multiple dualmode CPC (MDM-CPC): the flow direction of the liquid phases is altered by switching a valve in the CPC system

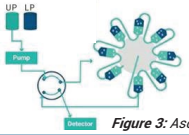


Figure 3: *Ascending mode separation*

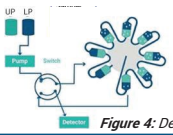
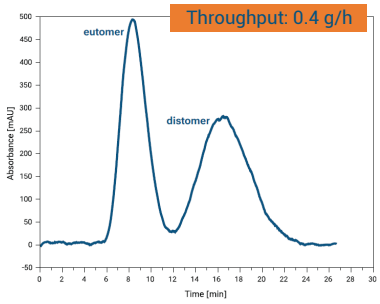


Figure 4: *Descending mode separation*

RESULTS & DISCUSSION

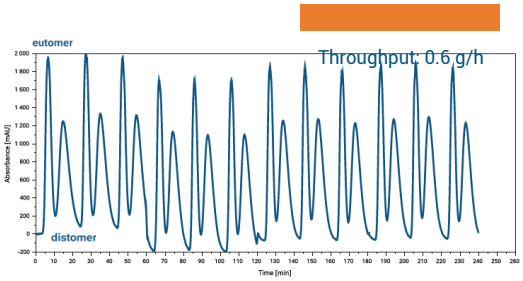
Batch Method



Solvent system	alkane/ester/SBEβCD$_{(aq)}$
Sample	200 mg in 20 mL UP
Flow	20 mL/min
Rotational speed	2500 rpm
Mode	asc/elution
Elution	100% UP

Figure 5: *Separation by CPC, batch mode*

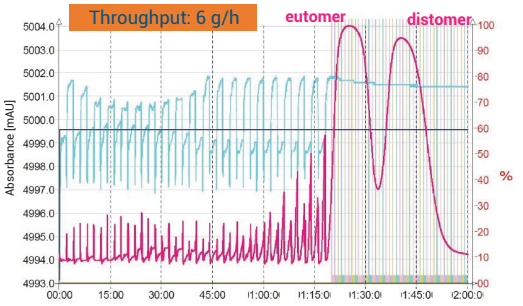
Stacked Injection Method



Total amount of sample	2.4 g/240 mL UP
Amount of sample/injection	200 mg/20mL
Flow	20 mL/min
Rotational speed	2500 rpm
Mode	asc/elution
Purity	99.1%
Yield	95%

Figure 6: *Separation by CPC, 12 consecutive stacked injections*

Multiple Dual-mode (MDM)



Total amount of sample	3 g/150 mL UP
Solvent system	Alkane/ester/MeβCD$_{(aq)}$
Flow	20 mL/min
Rotational speed	1800 rpm
Mode	multiple dual (ASC-DSC-ASC)
Purity	99.5+%
Yield	92%

Figure 7: *Separation by CPC, MDM continuous mode*

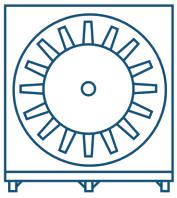
	Resolution	CPC – Stacked Injection	CPC - Multiple Dualmode (MDM)
Yield (%)	37	95	92
Mode	batch	Semi-batch	Continuous
Time of Process	2 days	Semi-batch	Continuous
Productivity (g/L*h)	0.28	0.80	5.0

Figure 8: *Comparative table between traditional resolution and continuous CPC methods*

CONCLUSIONS

Utilizing continuous CPC methods, we could achieve:

- **High productivity** – results almost in a 15x increase in productivity compared to conventional resolution
- **High valuable component yield**
- **Robust** and **reproducible** method
- **Cost efficient purification** – no need for expensive solid stationary phase or resolution agent , thereby reducing operational costs
- **Green purification method** – the cyclodextrine phase can be recycled and reused
- **Time efficiency** - CPC runs can be finished in much less time than traditional chromatographic steps
- **No Further Manipulation Needed** – The API will readily be available right after purification



REFERENCES

[1] Stephen James Dr. Ray, Kenneth Dr. Richardson, Triazole Antifungal Agents, European Patent Office, European Patent No. EP0440372A1, 1990
[2] Anita Kiss, Isomer Separation, RotaChrom Technologies PLC, Budapest, 2024

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